

Unit – 3 : Session – 11 : SLO – 1 | Integrate Event Handling in GUI Applications

I. Fill in the Blanks

1. ActionListener — the interface used to handle button clicks in Swing.
2. addActionListener — btn.addActionListener(listener) registers the listener.
3. getActionCommand() — returns the action command string from an ActionEvent.
4. java.awt.event — the ActionEvent class lives in the java.awt.event package.

II. True or False

1. TRUE — Swing dispatches events synchronously on the Event Dispatch Thread (EDT), so long tasks should be moved off it.
2. TRUE — one ActionListener instance can be registered with any number of buttons (distinguished via getActionCommand()).

III. Multiple Choice

1. B — actionPerformed(ActionEvent e) is the single method of the ActionListener interface.
2. B — an ActionEvent object is passed to actionPerformed().

IV. Programming

1. Output: C — clicking the button fires the lambda listener and prints "Hello".
2. Event-driven part: B — the anonymous ActionListener (its actionPerformed method) is the event-driven part of the snippet.